



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Discover. Learn. Empower.

EXPERIMENT :3

Student Name: Piyush Patel

UID: 23BCS12673

Branch: BE-CSE

Section/Group: KRG-1A

Date of Performance: 14/08/2025

Semester: V

Subject Name: ADBMS

Subject Code: 23CSP-333

1.Aim

Easy-Level Problem

Problem Title: Write and execute a SQL query that returns the maximum EMP_ID value from the Employee table excluding any EMP_IDs that appear more than once (i.e., only consider IDs that occur exactly once). Verify the result with the given sample data. **Objectives:**

- Understand grouping and filtering duplicate rows using GROUP BY and HAVING.
- Use a subquery in the WHERE clause to select only non-duplicated EMP_IDs.
- Return the maximum of those unique EMP_IDs using an aggregate function.

Expected Outcome:

- The query should return 7 for the provided sample data (since 7 is the largest EMP_ID that occurs exactly once).

Medium-Level Problem

Write and execute a SQL query to return the employee(s) with the highest salary in each department, along with their department name, using the given employee and department tables.

Objectives:

- Practice joining two related tables using INNER JOIN.
- Use correlated subqueries or aggregation (MAX) to find department-wise maximum salaries.
- Display employee name, salary, and corresponding department name.

Expected Outcome:

- For the given data, the result should show:
 - o Department IT: Employees JIM and MAX with salary 90000.
 - o Department SALES: Employee HENRY with salary 80000.

Hard-Level Problem

Given two tables TableA and TableB containing employee IDs, names, and salaries, write an SQL query to return each employee's ID, name, and their lowest salary across both tables.

Objectives:

- Understand how to combine data from multiple tables using UNION ALL.
- Use aggregation functions (MIN) to find the lowest salary per employee.
- Group results by employee ID and name to ensure correct mapping.

Expected Outcome:

- For the given data:
 - o Employee AA → Salary 1000
 - o Employee BB → Salary 300 (minimum of 300 and 400)
 - o Employee CC → Salary 100

2. Tools Used: SQL Server Management Studio

3. Query

Easy Level Problem

```
create table tbl_emp1(emp_id int);
```

```
insert into tbl_emp1(emp_id)
```

```
values(2),(4),(4),(6),(6),(7),(8),(8);
```

```
select * from tbl_emp1;
```

```
select max(emp_id) from tbl_emp1 where emp_id not in (select emp_id from tbl_emp1 group by  
emp_id having count(*) >1);
```

Medium Level Problem

```
CREATE TABLE department (
```

```
id INT PRIMARY KEY,
```

```
dept_name VARCHAR(50)
```

```
);
```

```
CREATE TABLE employee (
```

```
id INT,
```

```
name VARCHAR(50),
```



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Discover. Learn. Empower.

```
salary INT,  
    department_id INT,  
    FOREIGN KEY (department_id) REFERENCES department(id)  
);
```

```
INSERT INTO department (id, dept_name) VALUES  
(1, 'IT'),  
(2, 'SALES');
```

```
INSERT INTO employee (id, name, salary, department_id) VALUES  
(1, 'JOE', 70000, 1),  
(2, 'JIM', 90000, 1),  
(3, 'HENRY', 80000, 2),  
(4, 'SAM', 60000, 2),  
(5, 'MAX', 90000, 1);
```

```
select * from department;  
select * from employee;
```

--first approach

```
select D.dept_name, E.name, E.salary, D.id  
from employee as E  
join  
department as D  
on D.id=E.department_id  
where E.salary in  
( select max(E2.salary) from employee as E2 where E2.department_id=E.department_id )  
order by E.department_id;
```



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Discover. Learn. Empower.

--second approach

```
select D.dept_name, E.name, E.salary, D.id
from employee as E
join
department as D
on D.id=E.department_id
where E.salary in
( select max(E2.salary) from employee as E2 group by E2.department_id )
order by E.department_id;
```

Hard Level Problem

```
create table tableA(empId int , Ename varchar(50), salary int)
insert into tableA(empId, Ename, salary) values(1,'AA',1000), (2,'BB',300);
```

```
create table tableB(empId int , Ename varchar(50), salary int)
insert into tableB(empId, Ename, salary) values(2,'BB',400), (3,'CC',100);
```

```
select * from tableA;
select * from tableB;
```

--first approach

```
select empId, Ename, salary from tableA
UNION
select empId, Ename, salary from tableB
where salary < ( select min(salary) from tableA);
```

--second approach

```
select empId, min(Ename), min(salary) as 'Salary' from
(
select * from tableA
```

UNION All

select * from tableB

) as temp_result

group by empId;

4. Output:

EASY:

120 % 1 0

Results Messages

	(No column name)
1	7

MEDIUM:

120 % 5 0

Results Messages

	dept_name	name	salary	id
1	IT	JIM	90000	1
2	IT	MAX	90000	1
3	SALES	HENRY	80000	2

HARD:

120 % 5 0

Results Messages

	empId	(No column name)	Salary
1	1	AA	1000
2	2	BB	300
3	3	CC	100



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Discover. Learn. Empower.

5. Learning Outcomes

- a) Learn to implement subqueries to get the required data and also learnt the order of execution of subqueries.
- b) Understand the use of subqueries in simplifying the complex SQL operations.